

Product Submission to the Therapeutic Advisory Service
Meeting Date: 24th September

Minutes / TAS decision

Members noted that both strengths have shown non-inferiority to insulin glargine but have the advantage of longer half-life allowing greater flexibility of dosing. Reduction in volume is possible using 200unit per ml – this is a licensed product whereas U500 which it would replace is not. Dr Gregory had provided detailed algorithms of place in therapy and review / stopping criteria.

Cost is significantly higher than glargine. Patent expiry of glargine due in 2014/15 Possibility of biosimilars in production.

TAS decision: Supported for use in the specific patient groups in the algorithm supplied

INFORMATION SUPPLIED BY REQUESTOR

Consultant	Dr Rob Gregory	Clinical Business Unit	Acute
Speciality	Metabolic medicine	Divisional Support for Funding	Pending
Drug Name Generic	Insulin degludec	Drug Name Proprietary	Tresiba®
Formulation and strength		Solution for injection containing 100 units or 200 units of insulin degludec per ml.	
Typical Maintenance dose		Approx 40 to 400 units per day	
Clinical Indication: Treatment of diabetes mellitus in adults			
Licensed Indication? Yes		Licensed product? Yes	
Are patients currently using this agent in UHL as part of a clinical trial? No		How many patients per year? 100 units per ml – 30 patients 200 units per ml – 30 initially then 20 per year.	

INFORMATION SUPPLIED BY MEDICINES INFORMATION**SUMMARY**

- New basal insulin with long half life available in higher strength formulation
- Phase III controlled trials have shown insulin degludec to be non-inferior to insulin glargine at controlling HbA_{1c} and in patients with type 2 diabetes incidence rate of severe hypoglycaemia was reduced
- Safety assessed over 104 weeks was in line with other long acting insulin preparations
- Insulin degludec is more expensive than comparators glargine and detemir
- Long half life would allow greater dosing flexibility, and higher dose would allow lower volumes to be administered.
- Not approved by the FDA

COST COMPARISON * Per pack	Product	Manufacturer	Dose	Cost **
	Insulin degludec 100units/ml (Tresiba®)	Novo Nordisk Ltd	Variable	£72.00 for five 3ml pre-filled pens
	Insulin degludec 200units/ml (Tresiba®)	Novo Nordisk Ltd	Variable	£144.00 for five 3ml pre-filled pens
	Insulin glargine 100units/ml (Lantus®)	Sanofi	Variable	£41.50 for five 3ml pre-filled pens
	Insulin detemir 100units/ml (Levemir®)	Novo Nordisk Ltd	Variable	£42.00 for five 3ml pre-filled pens
	Humulin R 500units/ml	Eli Lilly USA (Unlicensed in UK)	Variable	£250.74 for one 20ml vial

* MIMS 02/09/13

**Hospital prices may vary

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Patent expiries of note: Insulin glargine expires in 2014

Current Status

TAS	Not previously reviewed
NICE	Not reviewed. Two evidence summaries on type 1 (ESNM24) and 2 (ESNM25) diabetes.
SMC	Not supported 856/13
Pharmaceutical Risk assessment	Low

Literature Search

Background

Insulin degludec (Tresiba®) is new long-acting basal insulin for the treatment of diabetes mellitus in adults. It is available in two strengths, 100 units per ml and 200 units per ml, making it the only licensed high strength insulin available in the UK.¹

Current Use (Local)

The Leicestershire Medicines Formulary includes insulin glargine as first choice, and insulin detemir is reserved for those patients unable to manage the insulin glargine device. Patients with type 1 diabetes should be offered an intermediate acting insulin first line; however a long-acting insulin analogue may be preferred in adolescents. An intermediate acting insulin is also considered first line for patients with type 2 diabetes requiring insulin. However long-acting insulin analogues may be considered where the patient requires assistance to inject and use the insulin, where hypoglycaemic episodes are affecting the patients lifestyle, where target HbA_{1c} is not reached with the intermediate insulin or where the patient would otherwise need twice daily intermediate acting insulin in combination with oral glucose-lowering drugs.² Some profoundly insulin resistant type 2 diabetics require large insulin doses to achieve glycaemic control, however large volumes of the standard insulin can cause considerable pain and discomfort on injection. One solution which has been used at UHL is the stronger unlicensed U500 insulin which contains 500 units per ml.³

Clinical Trials

(See Nice Evidence Summaries)

References

1. Insulin degludec for diabetes mellitus. Drug and therapeutics bulletin. 2013; 51(7):78-81
2. Leicestershire Medicines Formulary. Available at <http://netformulary.xuhl-tr.nhs.uk/netformulary/> accessed 03/09/2013
3. Crasto W., Jarvis J., Hackett E. et al. Insulin U-500 in severe insulin resistance in type 2 diabetes mellitus. Postgraduate Medical Journal. 2009; 85 (1002): 219-222

Kelly Anderson, Formulary Support Pharmacist
Leicestershire Medicines Information Service
Date 03/09/13

Additional Information supplied by requestor

What drug(s) if any would Tresiba® 100units/ml replace?

Lantus® (insulin glargine), or as an alternative to CSII (insulin pump).

What are the advantages of Tresiba® 100units/ml over existing treatments?

Tresiba® (insulin degludec) is the only ultra-long-acting analogue insulin preparation. Its duration of action is >24 hours after a single injection, and it shows low day-to-day variability of glucose-lowering effect within-subjects. It offers an option as once daily basal insulin in the following circumstances: Type 1 patients who fulfil NICE criteria for a long-acting basal insulin, but who cannot achieve their target HbA_{1c} because they experience nocturnal hypoglycaemia or morning hyperglycaemia while taking Lantus® (insulin glargine). The only current alternative is a trial of CSII (insulin pump) for patients with type 1 diabetes, an approach that is not acceptable to all eligible patients. (SEE ALGORITHM) Patients with type 1 and type 2 diabetes treated with a basal-bolus regimen based on once daily Lantus® (insulin glargine) who work variable shift patterns that prevents them injecting the basal insulin at the same time every day. The flexibility of dose interval offered by Tresiba® may deliver improved glucose control with variable

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dose intervals (minimum 8 hours). This property offers an alternative basal insulin for patients with type 1 diabetes whose failure to meet target HbA1c is in part attributable to forgetting to take a proportion of basal insulin doses. Type 1 or type 2 patients who need the assistance of a District Nurse to inject insulin, but who have unacceptably large variations in pre-injection capillary blood glucose concentration reflecting in part the variation in arrival time of the nurse.

What drug(s) if any would Tresiba® 200units/ml replace?

U500 Humulin R®

What are the advantages of Tresiba® 200units/ml over existing treatments?

It is the only basal higher strength insulin preparation available in the UK. It is administered with a pen injector that is calibrated for the strength of insulin – significantly reducing the risk associated with U500 Humulin R®. (Patients currently use U100 insulin syringes which administer 5x the stated number of units per injection.) Shelf life of 8 weeks vs. 28 days. Potential for less wastage according to dose required. Procurement, prescribing, and dispensing is standard rather than a unique process requiring additional pharmacy resources to administer for an unlicensed drug. As an alternative to unlicensed U500 Humulin R® in treating insulin resistant patients who require >200 units insulin per day to achieve adequate glycaemic control. This includes switching a proportion of those currently taking doses of U500 Humulin R® at the lower end of the dose range, and thereafter starting insulin resistant patients on this product before progressing to unlicensed U500 Humulin R®. Patients treated with U500 Humulin R® take it before meals, but it contributes to their basal and prandial insulin requirements. Tresiba® (insulin degludec) U200 would be used to provide the basal requirement with prandial doses of standard (U100) strength short-acting insulin.

Is Tresiba® being used by other centres in the country? Please give specific details:

Insulin Degludec formulary inclusions – June 2013 (NB This list includes approvals for U100 and for U200 strength preparations.)

Chelsea and Westminster Hospital NHS Foundation Trust: After insulin glargine for Type 1 patients experiencing nocturnal hypoglycaemia on other basal insulins. Type 1 patients awaiting pump therapy. Type 2 patients on large basal doses Type 2 patients on large basal doses who would benefit from an OD vs. BD dosing regimen. Type 2 patients on Humulin R 500units/mL who are still not achieving glycaemic control.

Hillingdon Hospital NHS Foundation Trust: Type 1 only in adults over 18years with eGFR over 30, not during pregnancy. To be started in secondary care only in: Those poorly compliant with OD basal regime (shift, lifestyle etc.). Those with frequent/severe hypoglycaemia (notably nocturnal) and for whom other insulin titration/intervention (i.e. CSII) has already been tried or is inappropriate.

Walsall Healthcare NHS Trust Joint formulary via the APC. Specialist initiation only. Patient types not specified.

Worcestershire Acute Hospitals NHS Trust Joint formulary via the APC. Specialist initiation by consultant, DSN and GPwSI. Patient types not specified.

West Suffolk NHS Foundation Trust Type 1 & Type 2 patients struggling to obtain glycaemic control with their basal insulin on the current pathway (Insulin Glargine/Insulin Detemir) and/or at risk of hypoglycaemia, in particular nocturnal hypoglycaemia. Patients who require high dose of basal insulin and would benefit from 200 units/mL high dose formulation.

Basildon and Thurrock University Hospitals NHS Foundation Trust Hard to control Type 1 patients Type 1 patients pre-pump.

Bedford Hospital NHS Trust Luton and Dunstable Hospital NHS Foundation Trust Type 1 patients with unresolved nocturnal hypoglycaemia, chaotic/psychiatric patients, allergy to Lantus/Levemir, uncontrolled at high doses. Amber status for secondary care consultant initiation or primary care DSN

South Tyneside NHS Foundation Trust Patient types not specified Pan-Mersey Area Prescribing Committee This APC covers: • Aintree University Hospital NHS Foundation Trust • St Helens and Knowsley Teaching Hospitals NHS Trust • Warrington and Halton Hospitals NHS Foundation Trust • The Royal Liverpool and Broadgreen University Hospitals NHS Trust Patients who are considered suitable for treatment with Tresiba can only be approved following a meeting of the patient's Consultant, GP and Prescribing Adviser from the Trust. On a named patient basis.

Bolton NHS Foundation Trust Uncontrolled Type 1 patients & high dose insulin glargine patients

Hampshire Hospitals NHS Foundation Trust University Hospital Southampton NHS Foundation Trust Degludec supported for specialist recommendation only in: • poorly controlled type 1 patients • recurrent DKA • as a licensed option to large doses of insulin instead of Humulin R 500units/mL • Type 1 patients who experience frequent hypoglycaemic events (more than 3 in a year) • Type 2 larger patients requiring more than 80 units per shot
United Lincolnshire Hospitals NHS Trust Patients with Type 1 Diabetes Mellitus who have recurrent admissions with diabetic ketoacidosis due to poor adherence. Patients with Type 1 Diabetes Mellitus with problematic or recurrent nocturnal hypoglycaemia on other long-acting analogues and are not suitable for insulin pump therapy. A small number of patients with Type 2 Diabetes Mellitus with significant insulin resistance who might otherwise need large doses of Humulin R 500 units/mL insulin

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DOI: Members of the Department have at various times received speaker fees, sponsored places at meetings, and educational support from NovoNordisk. Individual consultants serve or have served on its advisory board.

References provided:

1. Heller S, Buse J, Fisher M et al. Insulin degludec, an ultra-long-acting basal insulin, versus insulin glargine in basal-bolus treatment with mealtime insulin aspart in type 1 diabetes. *Lancet*. 2012; 379(985): 1489-1497
2. Ratner RE, Gough S, Mathieu C et al. Hypoglycaemia risk with insulin degludec compared with insulin glargine in type 2 and type 1 diabetes: a pre-planned meta-analysis of Phase 3 trials. *Diabetes, Obesity and Metabolism*, 2012 doi: 10.1111/dom.12032.

Pharmacoeconomic category

B – New Treatment more effective but more expensive than comparable therapy.

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Supporting Information – Insulin Degludec

1. Algorithm for prescription of 100 unit per ml
2. Algorithm for prescription of 200 unit per ml
3. Proposed SCA 200 unit per ml

1. Algorithm for prescription of 100 unit per ml

This applies to patients aged over 18 years with type 1 diabetes who are taking basal-bolus insulin regimen.

Q1 Is the patient using Lantus® (insulin glargine) as basal insulin?

N – This algorithm does not apply Y – Go to Q2

Q2 Has the patient attended a DAFNE course or received instruction in carbohydrate counting and insulin dose adjustment?

Y – Go to Q3 N – Arrange for the patient to receive this before proceeding.

Q3 Does the patient fulfil NICE criteria for using long-acting analogue insulin*?

N – The algorithm does not apply Y – Go to Q4

Q4 Does the patient have satisfactory diabetes control – ie acceptable HbA1c, infrequent hypoglycaemia?

Y – The algorithm does not apply N – Go to Q5

Q5 Has the patient tried CSII (insulin pump) treatment but has either stopped it or wishes to stop it?

Y – Offer a trial of Tresiba® (insulin degludec). Go to review N – Go to Q6

Q6 Does the patient want a trial of CSII (insulin pump) treatment?

Y – Arrange this N – Offer a trial of Tresiba® (insulin degludec). Go to review.

Review

Since Tresiba® (insulin degludec) is more expensive than Lantus® (insulin glargine) it is important to establish that it is delivering the improvement required of it.

After a 3 month trial the specialist team will assess the response to Tresiba® and decide whether to ask the GP to prescribe under the shared care agreement or to stop it.

If the reason for using it was to reduce nocturnal hypoglycaemia, the frequency and severity of hypoglycaemia must be documented before starting and after 3 months. If there has been no significant improvement then the Tresiba® should be stopped.

If the reason for prescribing it was to address variably high fasting glucose concentrations, the patient's glucose log (preferably on a memory meter) before and after 3 months should be scrutinised and if there has been no significant improvement then the Tresiba® should be stopped.

If failure to meet HbA1c target was part of the rationale to start Tresiba®, then a repeat measurement after 3 months should be compared with the initial value and a failure to improve should normally result in stopping Tresiba®.

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2. Algorithm for assessment of suitability for treatment with Tresiba® (insulin degludec) 200 units/mL

Q1. Is the patient 18 years or above?

N – This algorithm does not apply Y – Go to Q2

Q2. Does the patient have diabetes?

N – This algorithm does not apply Y – Go to Q3

Q3 Is the patient currently taking insulin?

N – This algorithm does not apply Y – Go to Q4

Q4 Is the patient taking more than 200 units per day?

N – This algorithm does not apply Y – Go to Q5

Q5 Does the patient show signs of insulin resistance – acanthosis nigricans and/or multiple skin tags?

N – Check patient's compliance with insulin doses prescribed. This algorithm may not apply.

Y – The patient is insulin resistant – go to Q6

Q6 Is the patient taking standard strength (U100) insulin?

Y – Go to Q7 N.- Go to Q9

Q7 – Does the patient have acceptable diabetes control on their current treatment regimen?

Y – A change of treatment is probably not indicated at the moment. N – Go to Q8

Q8 – Have all reasonable attempts been made to address the patient's insulin resistance (modifications to diet, physical activity, medication including where appropriate GLP1 mimetics, pioglitazone and bariatric surgery)?

N – Consider these options before returning to Q8.

Y – The patient is probably eligible for a trial of Tresiba® (insulin degludec) 200 units/mL. Go to 'review'.

Q9 Is the patient taking U500 Humulin R® insulin?

N – This algorithm does not apply Y – Go to Q10

Q10 Is the patient taking more than 480 units per day?

Y – The dose is high enough to suggest that a trial of Tresiba® (insulin degludec) 200 units/mL is unlikely to be beneficial.

N - The patient is probably eligible for a trial of Tresiba® (insulin degludec) 200 units/mL. Go to 'review'.

Review

Treatment with Tresiba® (insulin degludec) 200 units/mL is considerably more expensive dose-for-dose than treatment with standard strength (U100) insulins, and even more expensive than treatment with unlicensed U500 Humulin R® insulin. Patients should be monitored carefully during the first 3 months of treatment to assess whether the trial of treatment has been successful, and whether continued prescribing under a shared care agreement is justified. Criteria for success will be improvement in diabetes control as measured by HbA1c, stability of diabetes control as indicated by acceptable frequency and severity of hypoglycaemia, and absence of side-effects. Weight gain is not a stop criterion.

3. Tresiba® (insulin degludec) U200 Shared Care Guideline (draft)

Tresiba® (insulin degludec) U200 for patients with severe insulin resistance

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Scope

Prescribing by General Practitioners

Purpose

To provide advice on the safe prescribing of **Tresiba® U200** insulin for patients with severe insulin resistance.

Introduction

In the UK insulin is usually formulated at a concentration of 100 units per ml. **Tresiba® U200** insulin is a concentrated formulation of insulin (200 units per ml).

Patients with severe insulin resistance may have significantly higher insulin dose requirements than other patients with diabetes. The insulin dose can often reach more than 200 and up to 2000 units per day. Each insulin dose then requires two or three injections as the maximum insulin that can be injected in one injection is about 60-80 units. Larger volumes injected subcutaneously may affect the pharmacokinetics of the insulin. Large volume and multiple injections can also cause discomfort, leading to poor compliance and poor glycaemic control. Hitherto the only available higher strength insulin has been **U500 Humulin R®** (500 units/mL), which is only available in 20 ml vials, and has to be administered using standard U100 insulin syringes, calibrated to deliver one fifth of the true dose. This is unsatisfactory at best, and dangerous at worst.

Tresiba® U200 insulin should be considered for patients who require large doses of insulin (usually above 200 units per day), as this can reduce the volume of injected insulin, improve compliance, and can lead to improvements in glycaemic control by improvement in absorption.

Shared care

Sharing of care assumes communication between the specialist, GP and the patient. The intention to share care should be explained to the patient and accepted by them. Patients are under regular follow-up and this provides an opportunity to discuss drug therapy. The doctor who prescribes the medication has the clinical responsibility for the drug and the consequences of its use.

Shared care responsibilities

5.1 Consultant

The decision to start **Tresiba® U200** insulin will be made by a Consultant Diabetologist providing specialist care. A copy of the shared care guideline will be sent to the GP. If the GP has any queries about the prescribing of **Tresiba® U200**, they should contact the relevant hospital specialist as soon as possible.

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The consultant and the specialist team will:

- Educate the patient fully regarding safe administration, dosing schedule and storage of **Tresiba® U200** insulin, and provide guidance to the patient on appropriate blood glucose monitoring.
- Provide the patient with written information regarding safe administration, dosing schedule and storage of **Tresiba® U200** insulin and guidance in dealing with hypoglycaemia, concurrent illnesses and hospital admission.
- Send a letter to the GP requesting shared care for this patient.
- Provide specialist clinic follow-up on a regular basis.
- Send a letter to the GP after each clinic attendance documenting current dose and most recent blood results.
- Evaluate any reported adverse effects by GP or patient.
- Advise GP on review, duration or discontinuation of treatment with **Tresiba® U200** insulin where necessary.
- Inform GP of patients who do not attend clinic appointments.
- Ensure that backup advice for the patient and GP is available.

Hospital Pharmacy

- Provide initial supplies of **Tresiba® U200** for a maximum of 3 months.

5.2 General Practitioner

- Monitor patient's overall health and well being.
- Prescribe the **Tresiba® U200** insulin.
- Prescribe blood glucose monitoring strips as required.
- Report any adverse events to the hospital specialist, where appropriate.

5.3 Patient

- Discuss potential benefits and side effects of treatment with the specialist and GP, to identify whether they have a clear picture of these from the specialist and to raise any queries.
- Check that the specialists have provided a patient-held record or information sheet to alert other clinical staff to the treatment they are receiving.
- Share any concerns they have in relation to treatment with **Tresiba® U200** insulin.
- Report any adverse effects to their specialist or GP whilst taking **Tresiba® U200** insulin.
- Report to the specialist or GP if they do not have a clear understanding of their treatment.
- Participate in the monitoring of blood glucose, to assist health professionals to provide effective, safe, appropriate treatment.
- Inform relevant health professionals of the use of **Tresiba® U200** insulin at every contact and during any admission to hospital.

Dose and administration

Tresiba® U200 insulin is a long-acting insulin analogue synthesized by rDNA technology in *Saccharomyces cerevisiae*. It is a basal insulin for once daily administration, preferably at the same time each day.

It binds specifically to the human insulin receptor and has the same actions as human insulin.

Upon subcutaneous injection it forms soluble multi-hexamers resulting in a depot from which insulin degludec is continuously and slowly absorbed into the circulation leading to a flat and stable glucose lowering effect over 24 hours. The duration of action is >42 hours in the therapeutic dose range. Steady state will occur after 2-3 days of administration.

Tresiba® U200 insulin is for subcutaneous injection only. It must not be used intravenously or intramuscularly, or in insulin infusion pumps.

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Tresiba® U200 insulin comes in a pre-filled pen (FlexTouch) designed to be used with NovoFine or NovoTwist needles. The pen delivers 2-160 units per injection, in steps of 2 units. The dose counter shows the number of units and no dose conversion is required.

Tresiba® U200 insulin will be commenced by specialist clinic at University Hospitals of Leicester NHS Trust. The Clinic will provide the prescriptions for the first three months and the GP will provide prescriptions thereafter. The starting dose will be decided on the basis of the doses of insulin already being taken by the patient. This will include patients currently taking high doses of 100 units/mL insulin preparations, and selected patients taking **Humulin R® U500** insulin (500 units/mL), an unlicensed medication in the UK. The doses will be titrated according to the regular blood glucose monitoring performed by the patient and by the presence/absence of hypoglycaemia. The patient will be stabilized on **Tresiba® U200** insulin by the specialist team.

The patient will be given education regarding the safe storage and administration of the insulin by the specialist team.

The patient will be issued with a treatment card with details of their insulin and doses and instructions to be followed in case of illness, emergency, hypoglycaemia or hospital admission.

Adverse Effects

Hypoglycaemia

As with all insulins, the most frequent adverse event is hypoglycaemia. As with all basal insulin products, the prolonged effect of **Tresiba® U200** may delay recovery from hypoglycaemia.

Symptoms of mild to moderate hypoglycaemia may occur suddenly and can include:

- sweating • drowsiness
- dizziness • sleep disturbances
- palpitations • anxiety
- tremor • blurred vision
- hunger • slurred speech
- restlessness • depressed mood
- tingling in the hands, feet, lips, or tongue • irritability
- light headedness • abnormal behaviour
- inability to concentrate • unsteady movement
- headache • personality changes

Signs of severe hypoglycaemia can include:

- disorientation • seizures
- unconsciousness • coma
- death

Early warning symptoms of hypoglycaemia may be different or less pronounced under certain conditions, such as long duration of diabetes, autonomic diabetic neuropathy, use of medications such as beta blockers, changing insulin preparations, or intensified control of diabetes. Without recognition of early warning symptoms, the patient may not be able to take steps to avoid more serious hypoglycaemia. Patients who experience hypoglycaemia without early warning symptoms should monitor their blood glucose more frequently, especially prior to activities such as driving.

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Mild to moderate hypoglycaemia may be treated by eating foods or taking drinks that contain sugar. Patients should always carry a quick source of sugar, such as fruit juice, non-diet carbohydrate-containing drinks or glucose tablets.

The patient will receive education by the specialist team regarding safe detection and management of hypoglycaemia.

Hypokalaemia

Insulin stimulates potassium movement into the cells, possibly leading to hypokalaemia, which if untreated may cause respiratory paralysis, ventricular arrhythmia, and death. Use caution in patients who may be at risk of hypokalaemia (e.g., patients using potassium-lowering medication e.g. thiazide diuretics).

Hypersensitivity and Allergic Reactions

Severe, life-threatening, generalized allergy, including anaphylaxis, can occur with insulin products, and this would be expected to apply to **Tresiba® U200** insulin. Localized reactions and generalized myalgias have been reported with the use of metacresol as an injectable excipient.

Local Allergy —Patients occasionally experience erythema, local oedema, and pruritus at the site of injection. This condition usually is self-limiting. In some instances, this condition may be related to factors other than insulin, such as irritants in the skin cleansing agent or poor injection technique.

Weight gain

Weight gain can occur with some insulin therapies and has been attributed to the anabolic effects of insulin and the decrease in glycosuria.

Peripheral Oedema

Insulin may cause sodium retention and oedema, particularly if previously poor metabolic control is improved by intensified insulin therapy.

Hyperglycaemia, Diabetic Ketoacidosis, and Hyperosmolar Non-Ketotic Syndrome

Hyperglycaemia, diabetic ketoacidosis, or hyperosmolar coma may rarely develop if the patient takes less **Tresiba® U200** insulin than needed to control blood glucose levels. This could be due to increases in insulin demand during illness or infection, neglect of diet, omission or improper administration of prescribed insulin doses or use of drugs that affect glucose metabolism or insulin sensitivity. Early signs of diabetic ketoacidosis include glycosuria and ketonuria. Polydipsia, polyuria, loss of appetite, fatigue, dry skin, abdominal pain, nausea and vomiting and compensatory tachypnoea come on gradually, usually over a period of some hours or days, in conjunction with hyperglycaemia and ketonaemia. Severe sustained hyperglycaemia may result in hyperosmolar coma or death.

Renal or Hepatic Impairment

Frequent glucose monitoring and insulin dose reduction may be required in patients with renal or hepatic impairment.

See the Summary of Product Characteristics (SPC) for further information on adverse events.

Monitoring Standards

Blood glucose monitoring	By the patient and reviewed by the specialist team and the GP
HbA1c/ blood glucose/U&E/eGFR lipid profile/FBC	Will be measured by hospital clinic initially – this may pass to the GP but only after consultation between the hospital specialist and the GP

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The specialist team will inform GPs and the patients of the frequency of blood glucose monitoring and HBA1c and other blood testing as required.

Actions to take in the event of abnormal test results/symptoms

If any unusual or serious adverse effect is reported please contact the specialist team for further advice.

Contact numbers for advice and support

Specialist	Post	Telephone
Specialist on call via Hospital Switchboard		
Dr Robert Gregory	Consultant	0116 2588017

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